

K Nearest Neighbours

It is a classification algorithm, based on distance metrics

where k is a number, it is called a hyper parameter

parameter: Provided by model

hyper parameter: Provided by user

analogy: Function we have some arguments, we can change function arguments

k = Number of neighbours

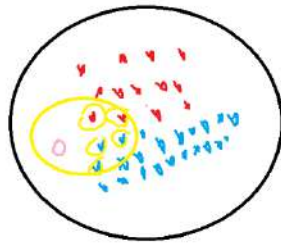
Hyper Parameter:

- *The parameters defined by user*
- *Decision Tree in order to find the root node, we have Gini, entropy*
- *user can chose which method wants to apply*
- *in LR To find the coefficients : OLS and Gradient descent*
- *will keep on change these hyper parameters to increasae the model performance*
 - *Hyper Parameter tuning*

Now onwards always keep an eye on each algorithm hyperparameters

- 1) *Linear regression*
- 2) *Logistic Regression*
- 3) *Elastic regression: L1 + L2*
- 4) *Naïve bayes*
- 5) *Decision tree*
- 6) *KNN*

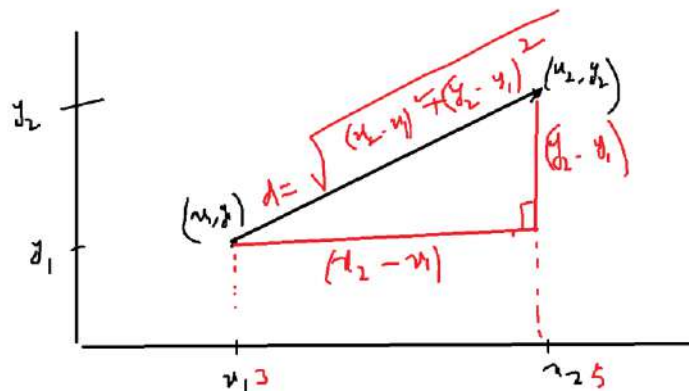
Which model is requires standardization, which model does not why?



In the KNN we know we need to calculate distance , there are many distance methods are there

- 1) Euclidian Distance
- 2) Cosine distance or cosine similiarity
- 3) Manhattan distance
- 4) Jacobian distance

Euclidian Distance



K NN : Nearest Neighbours

- *K is a hyper parameters , suppose $k = 3$*
- *we are choosing 3 neighbours*
- *The new data will calculate the distance between each neighbour*
- *which ever is the lowest distance ,it has more realtion*

$$D \propto \frac{1}{\text{Relation}}$$

Distance = Euclidain distance

$$(x_1, y_1) (x_2, y_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

We have from the people survey and objective testing with two attributes

x1=acid durability

X2=strength

to classify whether a special paper tissue is good or not.

X1=acid durability	X2=Strength	Y
7	7	Bad
7	4	Bad
3	4	Good
1	4	Good
3	7	?

Question:

If new tuple X1=3, and X2=7 what is Y?

(3,7) vs (7,7)

(3,7) vs (7,4)

(3,7) vs (3,4)

(3,7) vs (1,4)

Step – 1:

Calculate the distance between new tuple with all the data points

$$\text{Distance between } (3,7) \text{ vs } (7,7) = \sqrt{(7-3)^2 + (7-7)^2} = \sqrt{16} = 4$$

$$\text{Distance between } (3,7) \text{ vs } (7,4) = \sqrt{(7-3)^2 + (4-7)^2} = \sqrt{25} = 5$$

$$\text{Distance between } (3,7) \text{ vs } (3,4) = \sqrt{(3-3)^2 + (4-7)^2} = \sqrt{9} = 3$$

$$\text{Distance between } (x1:3,y1:7) \text{ vs } (x2:1,y2:4) = \sqrt{(1-3)^2 + (4-7)^2} = \sqrt{13} = 3.6$$

X1=acid durability	X2=Strength	Y	D
7	7	Bad	$\sqrt{16} = 4$
7	4	Bad	$\sqrt{25} = 5$
3	4	Good	$\sqrt{9} = 3$
1	4	Good	$\sqrt{13} = 3.6$

Step – 2:

Provide the ranking based on distances

Low distance : Rank high

X1=acid durability	X2=Strength	Y	D	Rank
7	7	Bad	$\sqrt{16}$	3
7	4	Bad	$\sqrt{25}$	4
3	4	Good	$\sqrt{9}$	1
1	4	Good	$\sqrt{13}$	2

Step – 3: Choose the neighbours (K)

- k – should always an odd number
- k can also find using similar to Elbow method(cluster analysis)
 - Error vs K graph
 - As K increases the error should decrease
 - At some point the error is constant choose that Optimal K
- $k = \sqrt{\text{no of observations}}$
- in this example $k = \sqrt{4} = 2 + 1 = 3$ or $2 - 1 = 1$
- In real time data will be huge, so neighbours also more

Why we are not choose $k = 1$

$K = 1$ model become overfit

$K = 1$ only one neighbour

The new observation always follows output of one neighbour

X1=acid durability	X2=Strength	Y	D	Rank	neighbour
7	7	Bad	$\sqrt{16}$	3	yes
7	4	Bad	$\sqrt{25}$	4	no
3	4	Good	$\sqrt{9}$	1	yes
1	4	Good	$\sqrt{13}$	2	yes

Step – 4: Provide neighbours original answer

X1=acid durability	X2=Strength	Y	D	Rank	neighbour	original
7	7	Bad	$\sqrt{16}$	3	yes	Bad
7	4	Bad	$\sqrt{25}$	4	no	-----
3	4	Good	$\sqrt{9}$	1	yes	Good
1	4	Good	$\sqrt{13}$	2	yes	Good

Step – 5: Conclusion

out of 3 neighbours , 2 are saying Good and One are saying Bad

Majority voting = Good

so for input ($x_1 = 3, x_2 = 7$) the output is $y = \text{good}$

Every classification model can use as Regression also

DT classifier === DT regression

KNN Classifier === KNN regression

If it is a regression :

Average of the Neighbours (KNN Imputer)

If it is a classification:

Majority voting

When KNN model will overfit:

Low bias – high variance

No pattern learning

at what value for model will overfit

For bias – variance trade of =====

model complexity incresaes === model become overfit

model complexity increase ===== more features

$K = 1$ model become overfit

$K = 1$ only one neighbour

The new observation always follows output of one neighbour

In bias – variance consider range = 0 to 1 only

x axis draw agianst $\frac{1}{k}$

x	$1/k$	
0.2	$\frac{1}{0.2} = 5$	low complex
1	$\frac{1}{1} = 1$	complex

